

$$\Rightarrow b = -2.$$

b_1, b_0	00	01	11	10
b_2	00	0	0	1
b_3	01	1	0	0
	11	0	0	0
	10	0	0	0

Handwritten annotations on the Karnaugh map:

- Group 1 (Red): Cells (0,0), (1,0), (5,0), (4,0) with index 13.
- Group 2 (Blue): Cells (0,1), (1,1), (5,1), (4,1) with index 14.
- Group 3 (Green): Cells (0,0), (0,1), (1,0), (1,1) with index 7.5.
- Group 4 (Green): Cells (4,0), (4,1), (5,0), (5,1) with index 16.

$$\Rightarrow \bar{b}_1 \bar{b}_0 + \bar{b}_2 \bar{b}_0 + b_3 \bar{b}_2 \bar{b}_1$$

$$1, 4, \overset{13}{\cancel{15}}, 16 \rightarrow G_1 - 1,$$

$$1, 5, 9, 13 \rightarrow G_2 - 2, \quad 15, 16 \rightarrow G_3 - 3$$